



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

AMETEK SINGAPORE PTE LTD
 (Compliance Test Solutions)
 43 Changi South Avenue 2, #04-01
 SINGAPORE 486164
 Frankie Chua Phone: 65 6484 2388

CALIBRATION

Valid To: December 31, 2017

Certificate Number: 2939.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ^{2,3} (±)	Comments
DC Voltage – Measure	(1 to 750) V > 750 V to 40 kV	0.5 %	HP 3458A, Brandenburg 139D HV meter
AC Voltage – Measure RMS 10 Hz to 50 kHz Peak and Peak-to-Peak 1 Hz to 1 GHz	(1 to 750) V 40 mV to 18 kV	0.6 % 1.8 %	HP 3458A LeCroy 7300A
DC Current – Measure	(0.1 to 1) A (1 to 25) A	0.5 % 1.1 %	HP 3458A shunt with DMM

Parameter/Equipment	Range	CMC ^{2,3} (±)	Comments
AC Current – Measure (10 Hz to 50 kHz)	(0.1 to 1) A (1 to 25) A	1.3 % 1.1 %	HP 3458A shunt with DMM
DC Resistance – Measure	1 mΩ to 750 MΩ	1 % + 0.3 mΩ	HP 3458 A
EFT Simulators – Voltage Rise Time Pulse Width Burst Duration/Period Repetition	 250 V to 8 kV ± 10 % 5 ns ± 30 % (35 to 150) ns (0.75 to 300) ms (5 to 100) kHz	 2.6 % 3 % 2.6 % 0.47 % 0.68 %	 IEC 61000-4-4 (2012), IEEE Std C62.41 (2002) Lecroy 7300A, Teseq CAS 3025
ESD Simulators – Contact Voltage Rise Time Peak Current T ₁ Current T ₂ Current	 (2 to 30) kV ± 5 % (0.6 to 1) ns (7.5 to 112.5) A ± 15 % (0.275 to 60) A ± 30 % (0.15 to 30) A ± 30 %	 0.9 % 8.3 % 3.6 % 3.6 % 3.6 %	 IEC 61000-4-2 (2008); ISO 10605 (2008/A1 2014) Brandenburg HV meter 139D, HP3458A, DMM, Lecroy 7300A, Teseq MD103 ESD target
PQT Simulators – Pulse Rise/Fall Time Phase Angle Voltage Measurement Time Interval Overshoot/Undershoot Inrush Current	 (1 to 5) μs (0 to 360)° ± 10° Up to 240 V _{AC} 1 ms to 1 s Less than 5 % (500 to 600) A	 4.7 % 0.23 % 0.054 % 1.2 % 1.8 % 1.6 %	 IEC 61000-4-11 (2004) Lecroy 7300A oscilloscope, Lecroy 625Zi oscilloscope, Teseq MD 200A HV probe, HILO shunt ISM 50/2, HP 3458A

Parameter/Equipment	Range ³	CMC ^{2, 3} (±)	Comments
Ring Wave Simulators –			
Rise Time			
Open Circuit	0.5 μ s $\pm$ 30 %	4.4 %	IEC 61000-4-12 (2006), IEEE Std. C62.41 (2002) Oscilloscope, HV probe, shunt
Short Circuit	(0.2 to 1) μ s	4.6 %	
Open Circuit Phase Shift	(0 to 360) $^{\circ}$ $\pm$ 10 $^{\circ}$	0.12 %	
Open Circuit Voltage	(0.2 to 6.6) kV $\pm$ 10 %	2.2 %	
Short Circuit Current	(6 to 550) A $\pm$ 10 %	2.2 %	
Oscillation Frequency	100 kHz $\pm$ 30 %	0.19 %	
Output Impedance	(12, 30) Ω $\pm$ 20%	3.1 %	
Decaying	(40 to 110) %	3.9 %	
Surge Simulators –			
Rise Time			
Open Circuit	(1 to 10) μ s $\pm$ 30 %	3.9 %	IEC 61000-4-5 (2005, 2014), IEEE Std C62.41 (2002)
Short Circuit	(1 to 8) μ s $\pm$ 20 %	1.9 %	
Pulse Width			
Open Circuit	(35 to 1000) μ s $\pm$ 20 %	2.2 %	Lecroy 7300A oscilloscope, Lecroy 625Zi oscilloscope, Teseq MD 200A HV probe, Pintek DP- 22kPro HV probe, HILO shunt ISM 50/2 shunt
Short Circuit	(10 to 320) μ s $\pm$ 20 %	1.6 %	
Open Circuit Phase Shift	(0 to 360) $^{\circ}$ $\pm$ 10 $^{\circ}$	0.13 %	
Open Circuit Voltage	500 V to 6.6 kV $\pm$ 10 %	2.2 %	
Short Circuit Current	80 A to 3.3 kA $\pm$ 20 %	1.6 %	

Parameter/Equipment	Range	CMC ^{2,3} (±)	Comments
Transient Generators – Transient Peak Voltage	10 V to 18 kV	1.8 %	ISO 7637 (2015), ISO 16750-2 (2012)
Rise/Fall Time	(500 to 800) ps > 800 ps to 1 ns (> 1 to 10) ns > 10 ns to 20 ms	8.5 % 5.6 % 4.7 % 2.2 %	Lecroy 7300A oscilloscope, Lecroy 625Zi oscilloscope, Lecroy HVP 120 HV probe, Teseq MD 200A HV probe, HILO ISM 50/2 shunt, RF attenuators/loads
Pulse Width	500 ps to 1 ns > 1 ns to 10 s	5.1 % 1.2 %	

II. Time & Frequency

Parameter/Equipment	Range	CMC ^{2,3} (±)	Comments
Frequency	10 Hz to 100 kHz	0.1 %	HP 3458A
	1 Hz to 1 GHz	1.2 %	LeCroy 7300A

¹ This laboratory offers commercial calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMC's represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ In the statement of CMC and indication of Range values, percentages are to be read as percent of reading or indication, unless noted otherwise.



Accredited Laboratory

A2LA has accredited

AMETEK SINGAPORE PTE LTD. (COMPLIANCE TEST SOLUTIONS) SINGAPORE

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSLI Z540-1-1994 and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 7th day of January 2016.

A handwritten signature in blue ink, appearing to read 'J. C. Burt'.

Senior Director of Quality & Communications
For the Accreditation Council
Certificate Number 2939.01
Valid to December 31, 2017

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.